

RFM and Cohort Analysis for Customer Segmentation and Retention

1. Introduction

This project implements RFM (Recency, Frequency, Monetary) Analysis and Cohort Analysis to segment customers based on purchasing behavior and evaluate retention trends. The objective is to identify high-value customers, detect churn risks, and provide actionable business insights.

2. Objectives

- Segment customers using RFM Analysis.
- Identify valuable and loyal customers.
- Detect customers likely to churn.
- Analyze retention through Cohort Analysis.
- Generate business insights.

3. Dataset Description

The dataset contains transaction records including CustomerID, InvoiceNo, InvoiceDate, Quantity, UnitPrice/Amount, and Sales.

4. Data Preprocessing

Duplicate records were removed, missing Customer IDs handled, date fields converted to datetime format, and sales values calculated. Additional features such as Invoice Month, Cohort Month, and Cohort Index were created.

5. Exploratory Data Analysis

EDA included monthly revenue trends, customer purchase distributions, transaction frequency analysis, and identification of top customers.

6. RFM Analysis

Recency measures how recently a customer purchased, Frequency measures how often they purchase, and Monetary measures total spending. Customers were scored using quintiles and assigned RFM scores.

7. Customer Segmentation

Customers were categorized into Champions, Loyal Customers, Potential Loyalists, At Risk Customers, and Others based on their RFM scores.

8. Cohort Analysis

Customers were grouped by acquisition month. Retention rates were calculated by tracking customer activity across subsequent months and visualized using retention matrices and heatmaps.

9. Results and Insights

Champions generated the highest revenue contribution. Loyal customers demonstrated repeat purchasing behavior. Most churn occurred shortly after the first purchase month, highlighting the importance of onboarding and retention campaigns.

10. Business Value

The solution helps businesses improve customer retention, reduce churn, optimize marketing spend, identify high-value customers, and increase customer lifetime value.

11. Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook.

12. Conclusion

Combining RFM and Cohort Analysis provides a comprehensive view of customer value and retention behavior, enabling data-driven business decisions.